The Concentration Engine, in Depth

Part III — from a scalar shock G to a signed distributional theorem

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This note rebuilds Part III block by block, derives the central result with every step shown, and then spends most of its length on the one place where the data overruled the original draft (the **U19 sign correction**). That correction is not cosmetic: it relocates the engine of rising top shares from the skill channel to the capital channel, and that relocation is what makes the unified framework internally consistent.

1 What the engine is for

The baseline escape engine (Part I) treats AI as a scalar productivity shifter G. That is enough to predict *growth* but says nothing about *who is paid*. The concentration engine reopens the production function and partitions inputs into those that AI augments and those it does not, so that a single shock G produces a growth effect *and* a distributional effect simultaneously. Concretely, its deliverable is the object

$$\gamma_1 \;=\; \frac{\partial S^P}{\partial G},$$

the marginal response of the top income share to the AI shock, which then enters the master decomposition

$$\frac{\partial g}{\partial G} \;=\; \underbrace{(\beta_2 + \beta_3 F)}_{\text{direct growth channel}} \;+\; \underbrace{\gamma_1\,\theta}_{\text{distributional feedback}}. \tag{1}$$

Everything below exists to pin down γ_1 *and its sign*. As we will see, γ_1 enters multiplied by θ, so a sign error in γ_1 propagates directly into the growth equation — which is exactly why U19 mattered.

2 The object: what "the top share" actually is

Income is paid to three factors: AI-embodied capital K, high-skill labour H, and low/mid-skill ("routine") labour L:

$$Y_{it} \;=\; w^H_{it} H_{it} + w^L_{it} L_{it} + R_{it} K_{it}.$$

The top income share is high-skill labour income plus the *elite-owned* slice of capital income:

$$S^P_{it} \;=\; \frac{w^H_{it} H_{it} + \kappa_{it}\, R_{it} K_{it}}{Y_{it}} \;=\; s_H + \kappa\, s_K. \tag{2}$$

The parameter $\kappa_{it} \in [0,1]$ is the fraction of capital income accruing to top earners. With dispersed share ownership κ is moderate; in an oligarchic transition economy it approaches 1. This is the seed of a *capital-concentration channel* that runs independently of the skill story — and, after U19, turns out to be the dominant one.

3 The assumption that does all the work

$$Y_{it} = \Big[\beta_K\,(A^K_t K_{it})^\rho + \beta_H\,(A^H_t H_{it})^\rho + \beta_L\,(A^L_t L_{it})^\rho\Big]^{1/\rho}, \qquad \rho = \frac{\varsigma - 1}{\varsigma}. \tag{3}$$

Two things are load-bearing, and every sign downstream depends on them.

1. **Gross substitutes: $\varsigma > 1 \Rightarrow \rho \in (0,1)$.** A firm swaps a back-office clerk for a model. If instead $\varsigma < 1$ (complements, $\rho < 0$), *every* result below flips. The whole engine therefore rests on AI being substitutable for routine labour; that claim deserves an explicit empirical defence in the paper.

2. **Augmentation pattern $A'_K > 0,\ A'_H > 0,\ A'_L \le 0$.** AI makes capital and cognitive skill more productive and does nothing for — or actively degrades — the marginal product of routine labour. This is skill-biased and capital-biased technical change in a single stroke.

   Throughout, write the *augmentation growth rates*

$$g_j \equiv \frac{A'_j}{A_j} = \frac{\mathrm{d}\ln A_j}{\mathrm{d}G}, \qquad j \in \{K, H, L\},$$

so the assumptions read $g_K > 0,\ g_H > 0,\ g_L \le 0$. The empirically relevant — and most consequential — ordering is

$$g_K \ \ge\ g_H \ >\ 0 \ \ge\ g_L, \tag{4}$$

i.e. capital is augmented at least as fast as skill. Keep (4) in view: it is the single fact that breaks the original draft.

4 Factor prices and the two forces in the skill premium

Each marginal product of (3) is $w_j = \partial Y/\partial X_j = \beta_j A_j^{\rho} X_j^{\rho-1} Y^{1-\rho}$, so the skill premium is

$$\frac{w_{it}^H}{w_{it}^L} = \frac{\beta_H}{\beta_L}\left(\frac{A_H}{A_L}\right)^{\rho}\left(\frac{H_{it}}{L_{it}}\right)^{\rho-1}. \tag{5}$$

Two channels move it:

- **Technology ratio $(A_H/A_L)^{\rho}$.** AI raises A_H and freezes/lowers A_L; since $\rho > 0$ the premium *rises with G*. Pure skill-bias.
- **Scarcity term $(H/L)^{\rho-1}$.** Because $\rho - 1 < 0$, a thin skilled workforce makes this term large: the premium per skilled worker is higher where skill is scarce.

Hold onto that last clause. *The premium per skilled worker is higher where skill is scarce, but the share going to skilled workers as a group is a different object.* That distinction is the entire resolution of the sign puzzle in §6.

5 The central theorem: income share is a tournament against the average

This is the heart of the engine. For a CES technology the factor income share equals

$$s_j = \frac{w_j X_j}{Y} = \frac{\beta_j (A_j X_j)^{\rho}}{\Sigma}, \qquad \Sigma \equiv \sum_k \beta_k (A_k X_k)^{\rho} = Y^{\rho}, \qquad \sum_j s_j = 1.$$

Take logs and differentiate in G with factor supplies fixed in the short run. The numerator gives

$$\frac{\partial}{\partial G}\ln\!\left[\beta_j (A_j X_j)^{\rho}\right] = \rho\,\frac{\partial \ln A_j}{\partial G} = \rho\, g_j,$$

and the denominator gives a *share-weighted average* of every factor's augmentation rate:

$$\frac{\partial \ln \Sigma}{\partial G} = \frac{1}{\Sigma}\sum_k \rho\,\beta_k (A_k X_k)^{\rho}\frac{A'_k}{A_k} = \rho \sum_k s_k g_k \equiv \rho\,\bar{a}.$$

Subtracting,

$$\boxed{\frac{\partial \ln s_j}{\partial G} = \rho\big(g_j - \bar{a}\big), \qquad \bar{a} \equiv \sum_k s_k\, g_k} \qquad \Longleftrightarrow \qquad \frac{\partial s_j}{\partial G} = \rho\, s_j\big(g_j - \bar{a}\big). \tag{6}$$

Read it as a race. A factor gains income share *iff* AI augments it faster than the economy-wide, share-weighted average rate $\bar{a}$. The hurdle $\bar{a}$ is dragged down by routine labour's term $s_L g_L \leq 0$. Capital and skill — augmented fastest — clear it; routine labour falls below it. The displaced share is taken *out of* s_L: literally out of the middle and bottom of the distribution.

Because $\sum_j s_j = 1$ and $\sum_j s_j g_j = \bar{a}$, the responses (6) sum to zero, $\sum_j \frac{\partial s_j}{\partial G} = \rho \sum_j s_j(g_j - \bar{a}) = 0$: the engine only ever *redistributes* shares, never creates them. That adding-up constraint is the accounting backbone of everything that follows.

5.1 Two robust signs and one fragile one

Rewrite the hurdle gap for each factor by pulling g_j out of $\bar{a}$:

$$g_j - \bar{a} = \sum_{k \neq j} s_k \, (g_j - g_k).$$

Under the ordering (4), $g_K = \max_j g_j$ and $g_L = \min_j g_j$, so:

$$g_K - \bar{a} = s_H(g_K - g_H) + s_L(g_K - g_L) \;>\; 0 \qquad \Rightarrow \quad \frac{\partial s_K}{\partial G} > 0 \quad (\textbf{always}), \tag{7}$$

$$g_L - \bar{a} = s_K(g_L - g_K) + s_H(g_L - g_H) \;<\; 0 \qquad \Rightarrow \quad \frac{\partial s_L}{\partial G} < 0 \quad (\textbf{always}), \tag{8}$$

$$g_H - \bar{a} = s_L(g_H - g_L) - s_K(g_K - g_H) \qquad \Rightarrow \quad \frac{\partial s_H}{\partial G} \quad \textbf{sign ambiguous.} \tag{9}$$

The capital share rises and the routine share falls *unconditionally*. The skill share is the only term whose sign is genuinely up for grabs — and (9) is precisely the line the original draft got wrong.

6 The sign puzzle and the U19 correction

6.1 Why a careful draft would still get (9) wrong

The intuitive story — "AI is skill-biased, so skilled labour gains share" — is exactly right in a *two-factor* world. Drop capital ($s_K = 0$, so $s_H + s_L = 1$). Then

$$g_H - \bar{a} = s_L(g_H - g_L) \;\geq\; 0 \qquad \text{always.}$$

With only skill and routine labour, skill-bias *mechanically* raises the skilled share; there is no room for any other sign. A note built on this two-factor intuition will write $\partial s_H / \partial G > 0$ and propagate a *positive* skill channel into γ_1. This is the natural origin of the U19 sign.

6.2 What the third factor does

Reintroduce capital and let (4) bite. From (9),

$$\frac{\partial s_H}{\partial G} \;=\; \rho \, s_H \Big[\underbrace{s_L(g_H - g_L)}_{\text{skill beats routine } (+)} \;-\; \underbrace{s_K(g_K - g_H)}_{\text{capital beats skill } (+)} \Big]. \tag{10}$$

Skilled labour now competes on two fronts. It *wins* against routine labour (first term, positive), but it *loses* against capital (second term, subtracted). The net sign is an empirical horse-race:

$$\frac{\partial s_H}{\partial G} > 0 \iff s_L(g_H - g_L) \;>\; s_K(g_K - g_H). \tag{11}$$

When capital is augmented much faster than skill ($g_K \gg g_H$) and the elite already hold a large capital share (s_K large), the right-hand side dominates and **the skilled share *falls*** — even while the skill *premium* per worker in (5) keeps rising. This is the puzzle dissolved: high pay per skilled head and a shrinking skilled-group share are fully consistent, because price per worker and aggregate share are different objects, and AI can push them in opposite directions.

The U19 correction. The original draft carried $\partial s_H/\partial G > 0$ (the two-factor intuition). The data — $g_K - g_H$ large, s_K large — put the economy on the wrong side of (11), so $\partial s_H/\partial G < 0$. The corrected sign does *not* threaten the headline result that top shares rise; it relocates the cause. Top shares rise through capital, not skill.

6.3 Why the headline survives: assembling γ_1

Differentiate the top share (2). Allowing κ to respond to G as well,

$$\gamma_1 = \frac{\partial S^P}{\partial G} = \underbrace{\rho\, s_H(g_H - \bar{a})}_{\text{skill channel (fragile)}} + \underbrace{\kappa\, \rho\, s_K(g_K - \bar{a})}_{\text{capital-concentration channel (robust } >0)} + \underbrace{s_K \frac{\partial \kappa}{\partial G}}_{\text{ownership channel}} . \qquad (12)$$

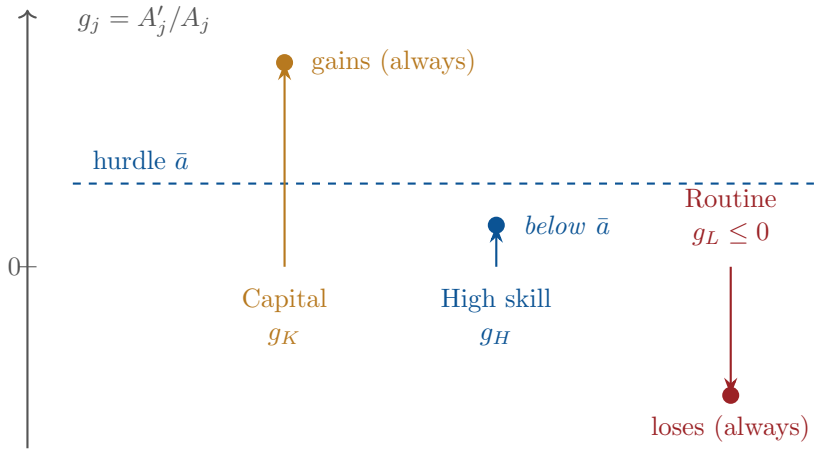
By (7) the middle term is strictly positive regardless of what skill does, and it scales with the elite ownership share κ. So:

- In an **oligarchic** economy ($\kappa \to 1$, s_K large), the capital term swamps a possibly negative skill term: $\gamma_1 > 0$ robustly.
- Only in the knife-edge of **fully dispersed** ownership ($\kappa \to 0$) does γ_1 reduce to the fragile skill term and risk turning negative.
- If AI also concentrates ownership ($\partial\kappa/\partial G > 0$, e.g. winner-take-all platform rents), the third term adds a further positive push.

The unified framework holds together because γ_1 rests on the *robust* leg (7), not on the fragile leg (9). The original draft happened to get $\gamma_1 > 0$ for the *wrong reason*; U19 keeps the conclusion and fixes the mechanism — which matters the moment anyone does comparative statics on skill supply or ownership policy.

7 The tournament, drawn

Vertical position is how fast AI augments each factor; the dashed line is the hurdle $\bar{a}$. Clear it and you gain income share; fall below it and your share is redistributed upward. Note that high-skill labour sits *below* the hurdle here — the U19 case.



In the original two-factor reading the hurdle sat between skill and routine and skill cleared it. Adding capital lifts $\bar{a}$ *above* g_H, and skill drops below the line.

8 A worked example: the sign flip in numbers

Take initial shares and augmentation rates consistent with (4), and $\varsigma = 2$ (so $\rho = \frac{1}{2}$):

$$(s_K, s_H, s_L) = (0.45,\ 0.20,\ 0.35), \qquad (g_K, g_H, g_L) = (0.080,\ 0.025,\ -0.020).$$

The hurdle is

$$\bar{a} = 0.45(0.080) + 0.20(0.025) + 0.35(-0.020) = 0.036 + 0.005 - 0.007 = 0.034.$$

Then, using (6) with $\rho = 0.5$:

Factor	g_j	$g_j - \bar{a}$	$\partial s_j/\partial G = \rho s_j(g_j - \bar{a})$	verdict
Capital K	0.080	+0.046	+0.01035	gains
High skill H	0.025	−0.009	−0.00090	**loses**
Routine L	−0.020	−0.054	−0.00945	loses
		sum	0.00000	(adds up)

Cross-check via (10): $s_L(g_H - g_L) - s_K(g_K - g_H) = 0.35(0.045) - 0.45(0.055) = 0.01575 - 0.02475 = -0.009$, matching the table. Skilled labour is on the losing side of (11).

Now feed the top share (12) (take $\partial\kappa/\partial G = 0$):

$$\gamma_1 = \underbrace{-0.00090}_{\text{skill}} + \kappa\,\underbrace{(0.01035)}_{\text{capital}}.$$

Ownership regime	κ	$\gamma_1 = \partial S^P/\partial G$
Oligarchic	0.80	$-0.00090 + 0.00828 = +0.00738$ (rises strongly)
Mixed	0.50	$-0.00090 + 0.00518 = +0.00428$ (rises)
Dispersed	0.20	$-0.00090 + 0.00207 = +0.00117$ (barely rises)
Fully dispersed	0.00	-0.00090 (**falls**)

The top share rises across every realistic ownership regime — and would *fall* only if the elite owned no disproportionate share of capital at all. That single table is the U19 result: the sign of γ_1 is governed by κ and the capital term, not by what happens to skilled wages.

9 Comparative statics worth stating in the paper

- **Skill supply does not rescue the middle.** Expanding H (more graduates) lowers the skill *premium* via $(H/L)^{\rho-1}$ in (5), but it does little to γ_1, which is anchored by the capital term. Education policy attacks the wrong channel if the goal is the top share.
- **Elasticity sign-flip.** If AI and routine labour are in fact complements ($\varsigma < 1$, $\rho < 0$), the hurdle logic inverts and the slowest-augmented factor *gains* share. The entire result is one elasticity sign away from reversing; §3 assumption (1) is the load-bearing empirical claim.
- **Ownership is policy.** Because γ_1 scales in κ, instruments that broaden capital ownership (sovereign wealth funds, broad equity holding, capital taxation feeding transfers) bear directly on the distributional term, whereas wage-side instruments do not.
- **Feedback into growth.** Through (1), $\gamma_1\theta$ now carries a robust, capital-driven sign. If $\theta < 0$ (concentration drags growth, e.g. via demand or political-economy frictions), the capital channel is also the growth-relevant one — the same parameter κ governs both inequality and its growth feedback.

10 Summary

The concentration engine turns a scalar shock G into a signed statement about the top share. The central theorem (6) makes income share a tournament against a share-weighted hurdle $\bar{a}$. Of the three contestants, capital wins and routine labour loses *unconditionally*; only high-skill labour's fate is ambiguous (9). The original draft inherited a two-factor intuition that forced the skilled share up; the data placed the economy on the other side of the horse-race (11), so U19 corrects $\partial s_H/\partial G$ to negative. The headline survives because γ_1 (12) rests on the robust capital term scaled by elite ownership κ, not on the fragile skill term. Top shares rise — through capital concentration, not the skill premium — and that is the sign the master decomposition (1) needs.